Bayes Theorem

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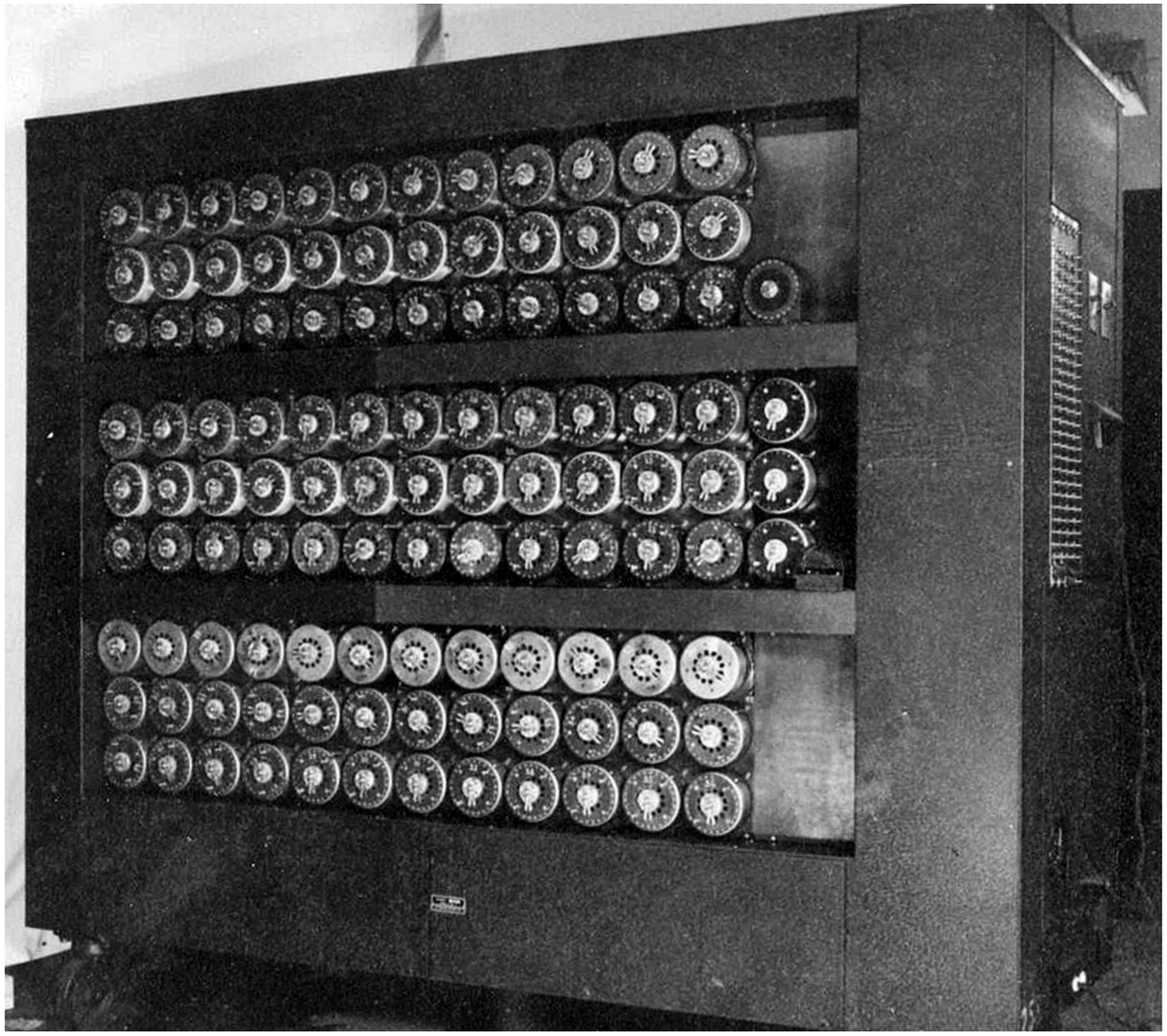
Credit to:

- Tom Hobbs (Colorado State University)
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What is the collective noun/term for a group of

- statisticians?
- Ecologists?





Wikipedia

Conditional Probability

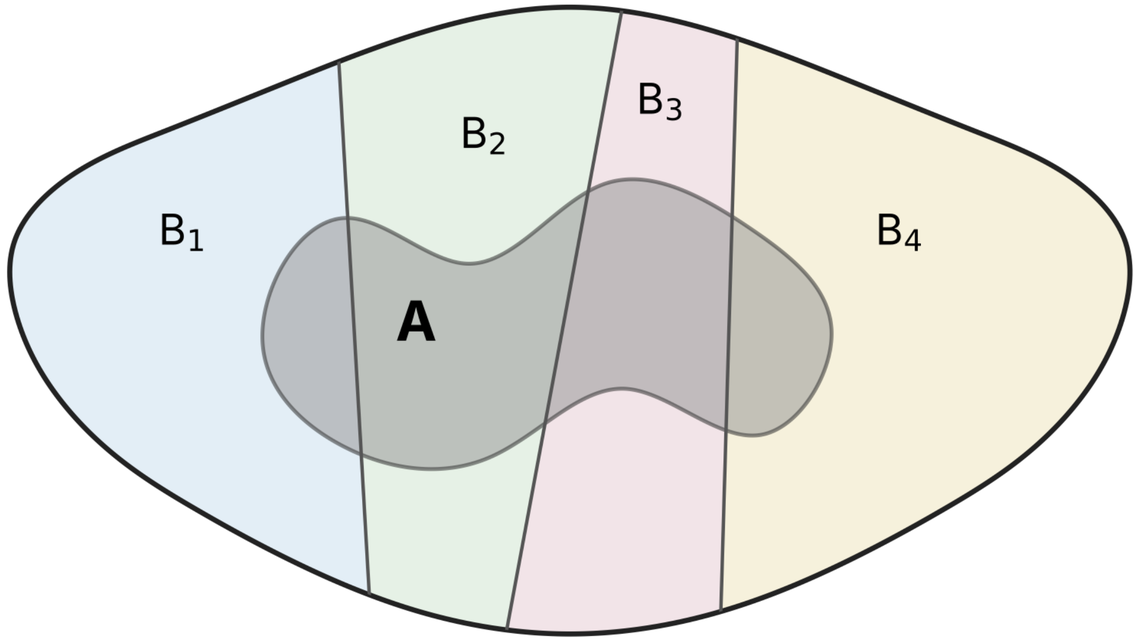
- When you have prior information and that can change the outcome
- Conditional Probability: Probability of event B happening given A is true
- Notation: [B|A] = [B,A] / [A]
- Probability of birds starting spring migration at night given winds are blowing towards north

- Notation: $[\theta|y] = [\theta,y] / [y]$
    - where θ is parameter (unknown) and y is Random Variable (Response Variable)

Law of Total Probability

- $[A] = [A \mid B_1][B_1] + [A \mid B_2][B_2] + [A \mid B_3][B_3] + [A \mid B_4][B_4]$
  - $[A] = \sum_i [A \mid B_i][B_i]$

- For continuous Random Variables
  - $[A] = \int_B [A|Bi][B]dB$



Why Bayes

- Bayes gives us the way to combine prior information with new data
- Frequentist approach gives a fixed value (the truth) for the response variable.
  - 95% Confidence Interval
- In Bayesian, there is “no truth”
  - Response variable does not have a fixed value but has some uncertainty to it - range of values that it can take.
  - 95% Credible Interval

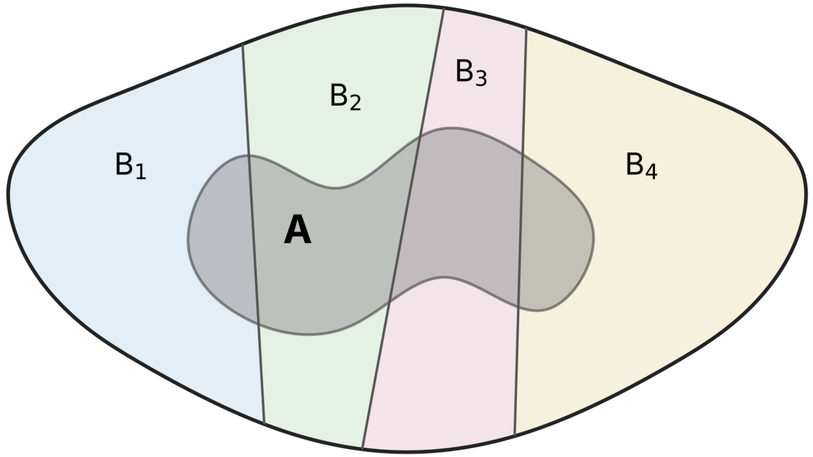
Bayes Theorem

- All unobserved quantities are Random Variables
- Remeber the conditional probability: $[\theta|y] = [\theta,y] / [y]$
- Bayes Theorem:

$$[\theta|y] = \frac{[y|\theta]\,[\theta]}{[y]}$$

- What is [y]?
- $[A] = \int_B [A|Bi][B]dB$ - Remember Law of Total Probability
  - Simialrly $[y] = \int [y|\theta]\,[\theta]d\theta$ for continuous parameters
- Thus, Bayes Theorem:

$$[\theta|y] = \frac{[y|\theta]\,[\theta]}{\int [y|\theta]\,[\theta]d\theta}$$

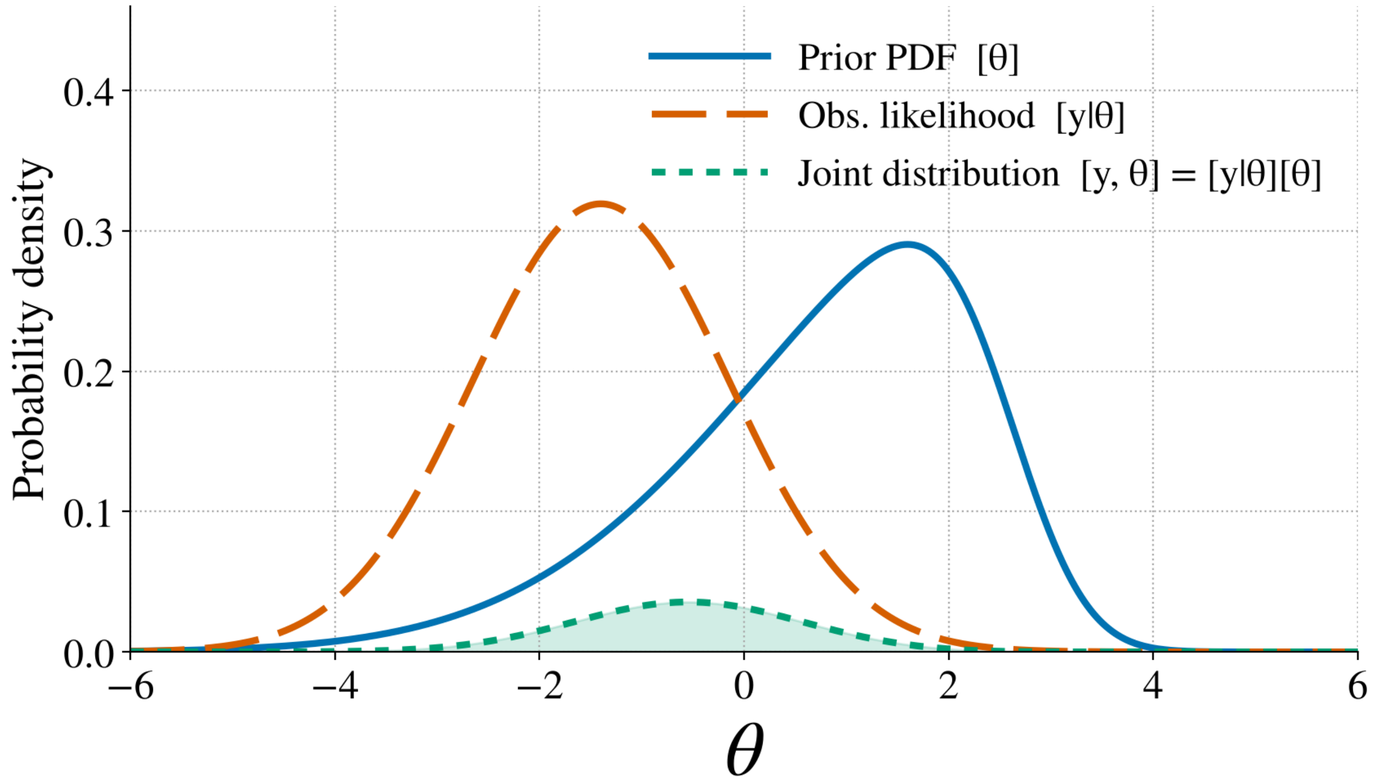


Bayes Theorem

- $[y] = \int [y|\theta]\,[\theta] d\theta$ is a constant after data is observed
  - It is also known as marginal distributions
  - Often impossible to calculate
- $[\theta|y] \propto [y|\theta]\,[\theta]$
  - $[\theta|y]$ - Posterior
  - $[y|\theta]$ - Liklihood
  - $[\theta]$ - Prior
  - $[y|\theta]\,[\theta]$ - Joint Distribution

Component of Bayes Theorem

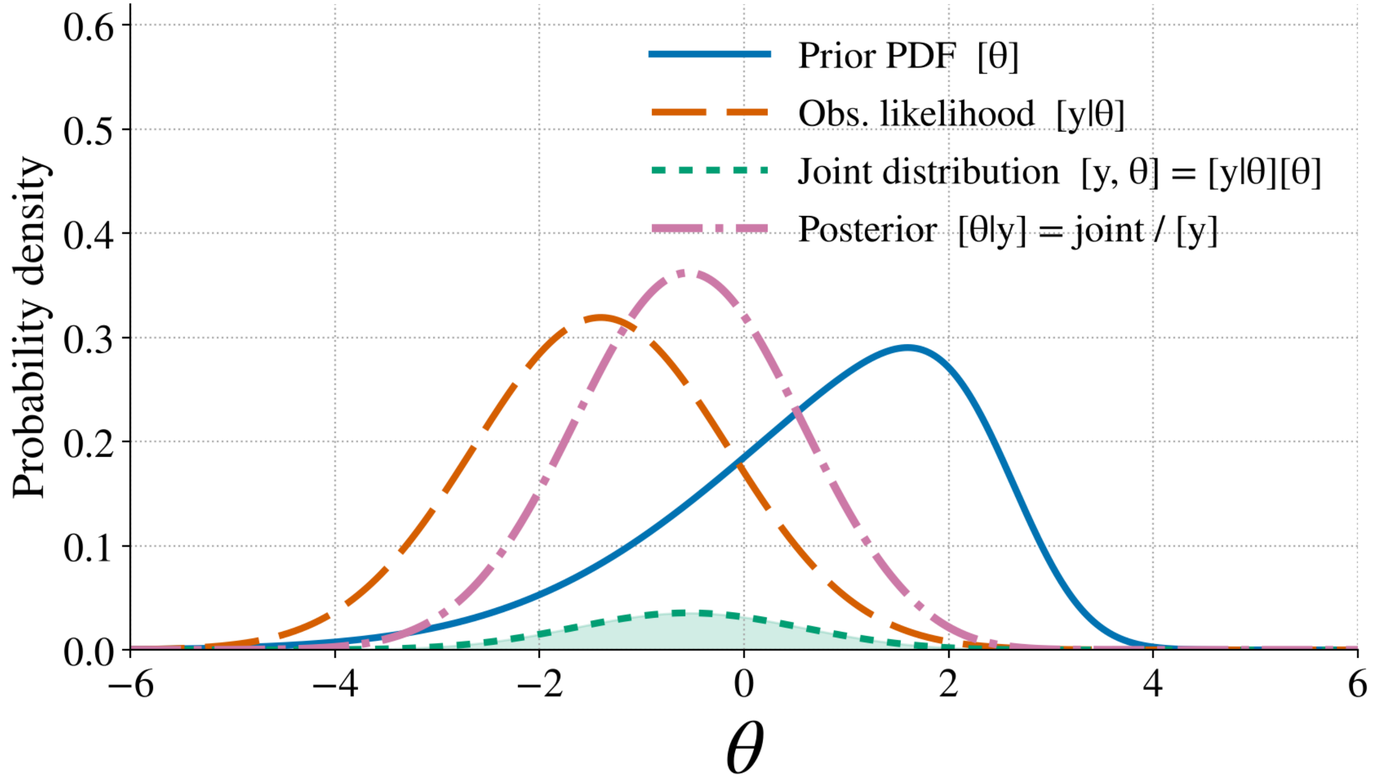
$$[\theta \mid y] = \frac{[y, \theta]}{[y]} = \frac{[y \mid \theta]\,[\theta]}{\int_\theta [y \mid \theta]\,[\theta]\,d\theta}$$



Shaded region - Marginal Distribution, it is the area under joint distribution

Component of Bayes Theorem

$$[\theta \mid y] = \frac{[y, \theta]}{[y]} = \frac{[y \mid \theta]\,[\theta]}{\int_\theta [y \mid \theta]\,[\theta]\,d\theta}$$

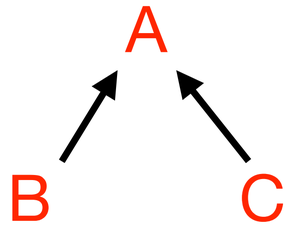


Shaded region - Marginal Distribution, it is the area under joint distribution

Convert this Theory to Application: Directed Acyclic Graphs Joint Distributions

Directed Acyclic Graphs: DAGs

- DAGs - Bayesian Network
  - How Joint distributions are factored into conditional distributions using nodes to represent Random Variables and arrows represent conditioning.
- Nodes at the heads of arrow should be on the left hand side of the conditioning symbol: “|”
- Nodes at the tail of arrow without an arrow leading to it must be expressed unconditionally.



[A|B,C] [B] [C]